Maximum Mode Timing Diagrams

Q. Discuss the Memory Read and Write timing diagrams for the 8086 microprocessor operating in Maximum Mode. How does it differ from Minimum Mode?

> **Definition to Remember**

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1. The Role of Status Signals

- **Pre-T1 Phase:** Before the T1 clock cycle even begins, the 8086 updates `S2', S1', S0'` to indicate the upcoming cycle (e.g., `101` for Memory Read, `110` for Memory Write).

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2. Memory Read Timing (Maximum Mode)

- **Pre-T1:** S2', S1', S0' = `1, 0, 1` (Memory Read).

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3. Memory Write Timing (Maximum Mode)

- **Pre-T1:** S2', S1', S0' = `1, 1, 0` (Memory Write).

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4. Key Differences from Minimum Mode

- **Signal Source:** 8288 generates ALE, DEN, DT/R', not the CPU.

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> **Must-Write Points (for 10 marks)**

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> **Quick Recall**

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